

Git & GitHub Complete Course (Basic to Advanced)

Module 1: Introduction to Version Control

Lesson 1

- What is Version Control?
 - Why Git?
 - History of Git
 - What is GitHub?
 - Git vs GitHub
 - Centralized vs Distributed Version Control
 - Advantages of Git
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Lesson 2

Installing Git

- Download Git
- Install on Windows
- Install on macOS
- Install on Linux

Verify Installation

```
git --version
```

Configure Git

```
git config --global user.name "Your Name"
```

```
git config --global user.email "you@example.com"
```

View Configuration

git config --list

Module 2: Git Basics

Lesson 3

Initialize Repository

git init

Check Status

git status

Create Repository

mkdir myproject
cd myproject
git init

Lesson 4

Git Workflow

- Working Directory
- Staging Area
- Local Repository
- Remote Repository

Add Files

git add file.txt
git add .

Commit Changes

git commit -m "Initial Commit"

Module 3: Viewing Changes

Lesson 5

View Commit History

`git log`

One Line History

`git log --oneline`

Difference Between Files

`git diff`

Show Commit Details

`git show`

Module 4: Undo Changes

Lesson 6

Restore File

`git restore filename`

Unstage File

`git restore --staged filename`

Reset Commit

`git reset`

Hard Reset

`git reset --hard HEAD`

Module 5: Branching

Lesson 7

Create Branch

`git branch feature`

View Branches

git branch

Switch Branch

git checkout feature

Modern Command

git switch feature

Lesson 8

Create & Switch

git checkout -b feature

Merge Branch

git merge feature

Delete Branch

git branch -d feature

Module 6: Merge Conflicts

Lesson 9

- What is Merge Conflict?
- Resolve Merge Conflict
- Best Practices

Practice

- Merge Two Branches
 - Resolve Conflict
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Module 7: Git Ignore

Lesson 10

Create

.gitignore

Ignore

node_modules/

*.log

.env

dist/

Module 8: Stashing

Lesson 11

Save Changes

git stash

View Stash

git stash list

Apply Stash

git stash apply

Delete Stash

git stash drop

Module 9: Tags

Lesson 12

Create Tag

`git tag v1.0`

Annotated Tag

`git tag -a v1.0 -m "Version 1.0"`

View Tags

`git tag`

Module 10: GitHub Introduction

Lesson 13

- Create GitHub Account
- GitHub Dashboard
- Repository
- Public vs Private Repository

Create Repository

Module 11: Connect Git to GitHub

Lesson 14

Add Remote

`git remote add origin https://github.com/user/project.git`

View Remote

`git remote -v`

Push Code

`git push -u origin main`

Module 12: Clone Repository

Lesson 15

Clone Repository

```
git clone https://github.com/user/project.git
```

Update Repository

```
git pull origin main
```

Download Latest

```
git fetch
```

Module 13: Push & Pull

Lesson 16

Push Changes

```
git push
```

Pull Changes

```
git pull
```

Fetch

```
git fetch
```

Module 14: GitHub Branches

Lesson 17

- Default Branch
- Feature Branch
- Protected Branch

Practice

Feature Development

Module 15: Pull Requests

Lesson 18

- Create Pull Request
 - Review Pull Request
 - Merge Pull Request
 - Close Pull Request
-

Module 16: Fork

Lesson 19

- Fork Repository
 - Clone Fork
 - Contribute to Open Source
-

Module 17: GitHub Issues

Lesson 20

- Create Issue
 - Assign Issue
 - Labels
 - Milestones
-

Module 18: GitHub Projects

Lesson 21

- Project Boards

- Kanban Board
 - Task Management
-

Module 19: Releases

Lesson 22

- Create Release
 - Release Notes
 - Download Release
-

Module 20: GitHub Pages

Lesson 23

Deploy Website

- Static HTML Website
 - Portfolio Website
-

Module 21: GitHub Actions

Lesson 24

Introduction

- Workflows
- CI/CD
- YAML Basics

Simple Workflow

Module 22: Collaboration

Lesson 25

Team Workflow

- Clone
 - Branch
 - Commit
 - Push
 - Pull Request
 - Merge
-

Module 23: Git Best Practices

Lesson 26

- Meaningful Commit Messages
 - Small Commits
 - Branch Naming
 - Code Review
 - Pull Before Push
-

Module 24: Advanced Git

Lesson 27

- Cherry Pick
- Rebase
- Squash Commits
- Amend Commit
- Reflog

Commands

`git cherry-pick`

`git rebase`

git commit --amend

git reflog

Module 25: Git Security

Lesson 28

- SSH Keys
 - Personal Access Token (PAT)
 - Two-Factor Authentication
 - Repository Permissions
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Module 26: Troubleshooting

Lesson 29

Common Errors

- Merge Conflicts
- Authentication Failed
- Detached HEAD
- Non-Fast-Forward
- Push Rejected

Solutions

Module 27: Final Projects

Lesson 30

Build Complete Git Workflow

Projects

- Student Management System

- Portfolio Website
 - E-commerce Website
 - Company Website
 - Team Collaboration Project
 - Open Source Contribution
 - GitHub Pages Portfolio
 - React Project Deployment
 - Node.js Project Version Control
 - Full Stack Project Repository
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30 Hands-on Practice Exercises

1. Install Git
 2. Configure Git
 3. Create Repository
 4. Add Files
 5. Commit Changes
 6. View Logs
 7. Create Branch
 8. Switch Branch
 9. Merge Branch
 10. Delete Branch
 11. Restore File
 12. Reset Commit
 13. Create Tag
 14. Stash Changes
 15. Create GitHub Repository
 16. Connect Local Repository
 17. Push Project
 18. Clone Repository
 19. Pull Updates
 20. Create Pull Request
 21. Resolve Merge Conflict
 22. Fork Repository
 23. Create Issue
 24. Create Release
 25. Deploy GitHub Pages
 26. Create GitHub Action
 27. Create SSH Key
 28. Rebase Branch
 29. Cherry Pick Commit
 30. Complete Team Collaboration Workflow
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Git & GitHub Interview Questions (Basic to Advanced)

Beginner

1. What is Git?
2. What is GitHub?
3. Difference between Git and GitHub?
4. What is Version Control?
5. What is a Repository?
6. What is a Commit?
7. What is the Staging Area?
8. What is a Branch?
9. What is a Merge?
10. What is `.gitignore`?

Intermediate

11. What is Git Workflow?
12. Difference between `git fetch` and `git pull`?
13. Difference between `git merge` and `git rebase`?
14. What is a Merge Conflict?
15. Explain Git Stash.
16. What are Tags?
17. What is a Pull Request?
18. What is Forking?
19. What is GitHub Pages?
20. What is GitHub Actions?

Advanced

21. What is Cherry Pick?
 22. Explain Rebase.
 23. What is Squashing Commits?
 24. What is Reflog?
 25. What is Detached HEAD?
 26. How do you recover lost commits?
 27. Explain SSH Authentication.
 28. Explain Personal Access Tokens (PAT).
 29. How do teams collaborate using Git?
 30. What are Git best practices?
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15 Real-World Mini Projects

1. Portfolio Website with GitHub Pages
 2. Student Management System
 3. Employee Management System
 4. Library Management System
 5. College Website
 6. Restaurant Website
 7. E-commerce Website
 8. Weather App Repository
 9. Todo App with Version Control
 10. Notes Application
 11. Blog Website
 12. React Portfolio
 13. Node.js REST API
 14. Full Stack MERN Project
 15. Team Collaboration Project
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Git Command Cheat Sheet

Repository Commands

```
git init
git clone <repository-url>
git status
git add .
git commit -m "Commit message"
git log
git log --oneline
```

Branch Commands

```
git branch
git branch feature
git switch feature
git checkout -b feature
git merge feature
git branch -d feature
```

Remote Commands

```
git remote -v
git remote add origin <repository-url>
git push -u origin main
git push
```

git pull
git fetch

Undo Commands

git restore file.txt
git restore --staged file.txt
git reset
git reset --hard HEAD
git revert <commit-id>

Advanced Commands

git stash
git stash pop
git tag
git cherry-pick
git rebase
git reflog